PROJECT DOCUMENTATION PHASE

DATE	**28 SEPTEMBER 2026**
TEAM ID	**6ab220cfbee0b08a2d2c63cd**
PROJECT NAME	**AUTO TICKET CLASSIFICATION USING FLOW DESIGNER**

1. INTRODUCTION

Project title:

Auto ticket classification using flow designer

Team members:

Nancy Mary S: project developer

Padmini P: project developer

Akil K: project developer

Muthu Sangeetha K: project developer

Muthuesakki S: project developer

2. PROJECT OVERVIEW

Purpose:

The auto ticket classification using flow designer project is developed to automate the classification of it support tickets in a school environment.

The school it helpdesk receives different types of complaints such as wi-fi problems, projector failures, password issues, and slow computer problems. Normally, it staff manually review each ticket and assign a category and subcategory.this project uses servicenow flow designer to automatically analyze the ticket description and classify the issue based on predefined keywords. This reduces manual effort and improves ticket routing efficiency.

Features:

- Automatic ticket classification.
- Automatic category assignment.
- Automatic subcategory assignment.
- Dependent category and subcategory fields.
- Automatic email notification.
- Auto-generated ticket number.
- No-code implementation using flow designer.
- Easy maintenance and scalability.

Classification Examples:

Keyword / Issue	Category	Subcategory
Wifi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

3. **ARCHITECTURE**

Frontend:

The project uses the servicenow platform interface for creating and submitting it support tickets.

The User Provides:

- Caller
- Short Description
- Description

The system then processes the ticket through flow designer.

Backend:

The Backend Functionality Is Implemented Using Servicenow Flow Designer.

The flow is triggered when a new record is created. It checks whether the category field is empty and then analyzes the short description. Based on the matching condition, the system updates the category and subcategory fields.

Backend flow:

New ticket

↓

Record created trigger

↓

Check category is empty

↓

Analyze short description

↓

If / else if conditions

↓

Update category & subcategory

↓

Send email

Database:

The project uses a custom servicenow table named: Incident workflow

The table stores ticket records and contains fields such as:

- Number
- Caller
- Category
- Subcategory
- Short description

Description:

- State
- Assigned group
- Assigned to

Database classification:

Field	type
Number	auto number
Caller	reference
Category	choice
Subcategory	choice
Short description	string

Description	string
State	choice
Assigned group	reference
Assigned to	reference

4. SETUP INSTRUCTIONS

Prerequisites:

The following are required:

- Servicenow account/instance
- Web browser
- Access to flow designer
- Access to system definition
- Access to update sets
- Installation / configuration

Step 1: create update set

Open all → system update sets → local update sets.

Click new.

Name it project update set.

Set state as in progress.

Set application as global.

Save.

Select make this my current set.

Step 2: create custom table

Create a table named: Incident workflow

Enable auto number and create the required fields.

Step 3: configure fields

Create the required fields such as category, subcategory, short description, description and state.

Step 4: configure dependency

Make subcategory dependent on category.

Examples:

Network → wi-fi

Hardware → projector

Access → forgot password

Performance → slow computer

Step 5: create flow

Create a new flow named: Auto classify school it tickets

Application: global.

5. FOLDER STRUCTURE

Since this project is developed in servicenow there is no traditional client and server folder structure.

The project structure can be represented as:

Auto ticket classification

```
│
├── update set
│   └── project update set
│
├── custom table
│   └── incident workflow
│
├── fields
│   ├── number
│   ├── caller
│   ├── category
│   ├── subcategory
│   ├── short description
│   ├── description
│   ├── state
│   ├── assigned group
│   └── assigned to
│
├── flow designer
│   └── auto classify school it tickets
│
```

└── email notification

└── ticket confirmation

6. RUNNING THE APPLICATION

The application is executed within the servicenow instance.

Process:

Login to servicenow.

Open the incident workflow table.

Create a new ticket.

Enter the caller.

Enter the short description.

Save the record.

Flow designer is triggered automatically.

The system identifies the issue.

Category and subcategory are automatically populated.

Email confirmation is sent to the caller.

The flow uses record created as its trigger and checks whether category is empty.

7. API DOCUMENTATION

API status:

This project does not use custom rest apis or external apis.The automation is implemented directly using servicenow flow designer.

Flow actions:

Action	purpose
Record created	starts the flow
If condition	checks ticket description
Update record	updates category/subcategory
Send email	sends ticket confirmation

Example:

Input:

Short description: Wifi not working in library

Output:

Category: network

Subcategory: wi-fi

8. AUTHENTICATION

The project uses the authentication and access-control mechanism provided by the servicenow platform.

Users must access the servicenow instance through authorized accounts.The project also validates that only authorized users can access or modify ticket data.

9. USER INTERFACE

The user interface is provided through the servicenow ticket form.

Main ui components

Caller field

Short description field

Description field

Category field

Subcategory field

State field

Assigned group

Assigned to

10. TESTING

The project was tested using different types of it support tickets**.**

Test case 1 – wi-fi issue

Input:

Wifi not working in library

Expected result:

Category → network

Subcategory → wi-fi

Actual result:

Passed.An email was also sent to the caller.

Test case 2 – projector issue

Input:

Projector not turning on

Expected result:

Category → hardware

Subcategory → projector

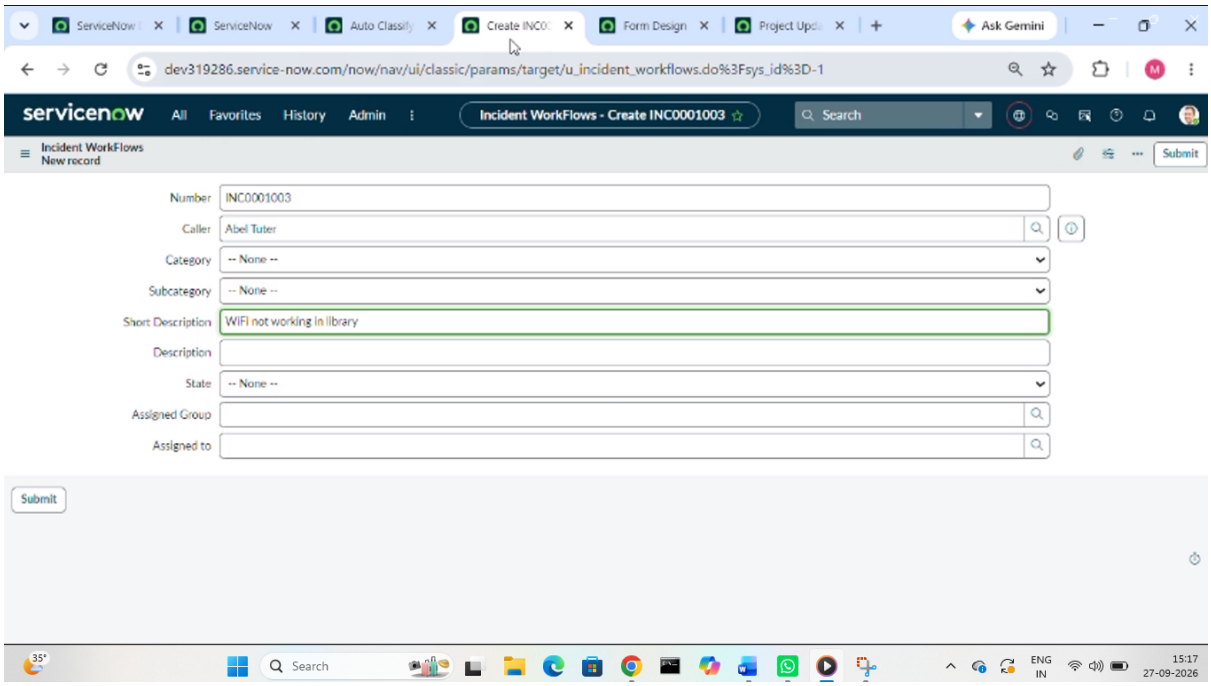
Actual result:

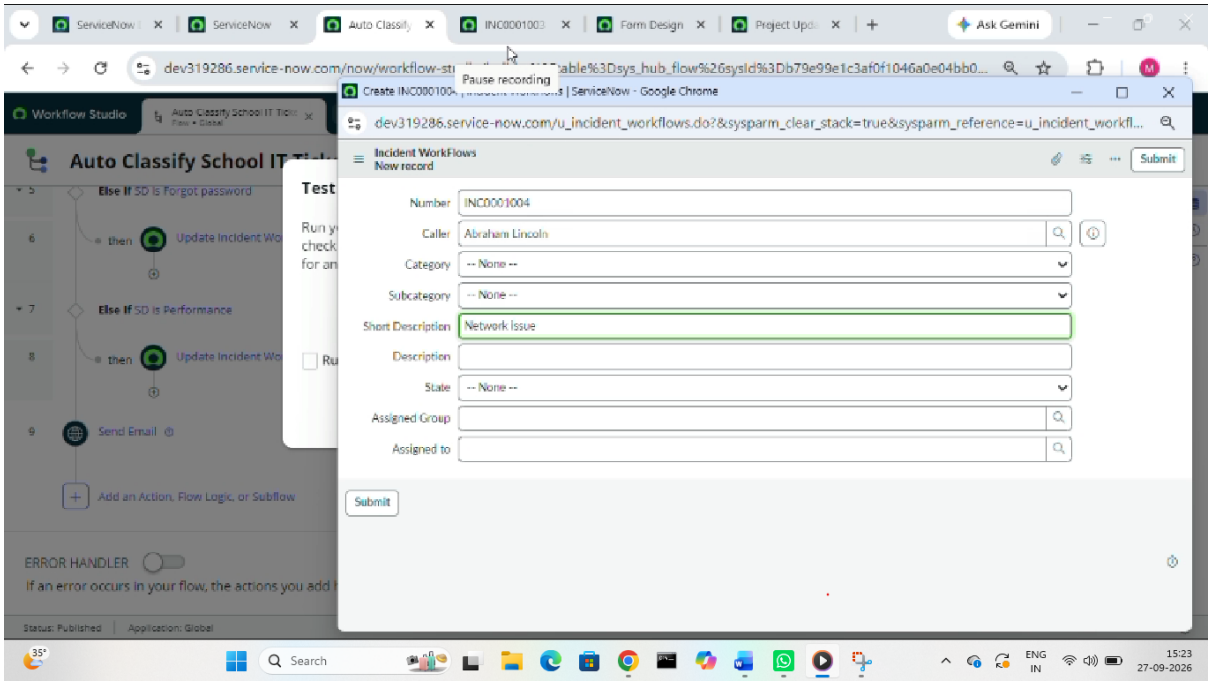
Passed.An email was sent to the caller.

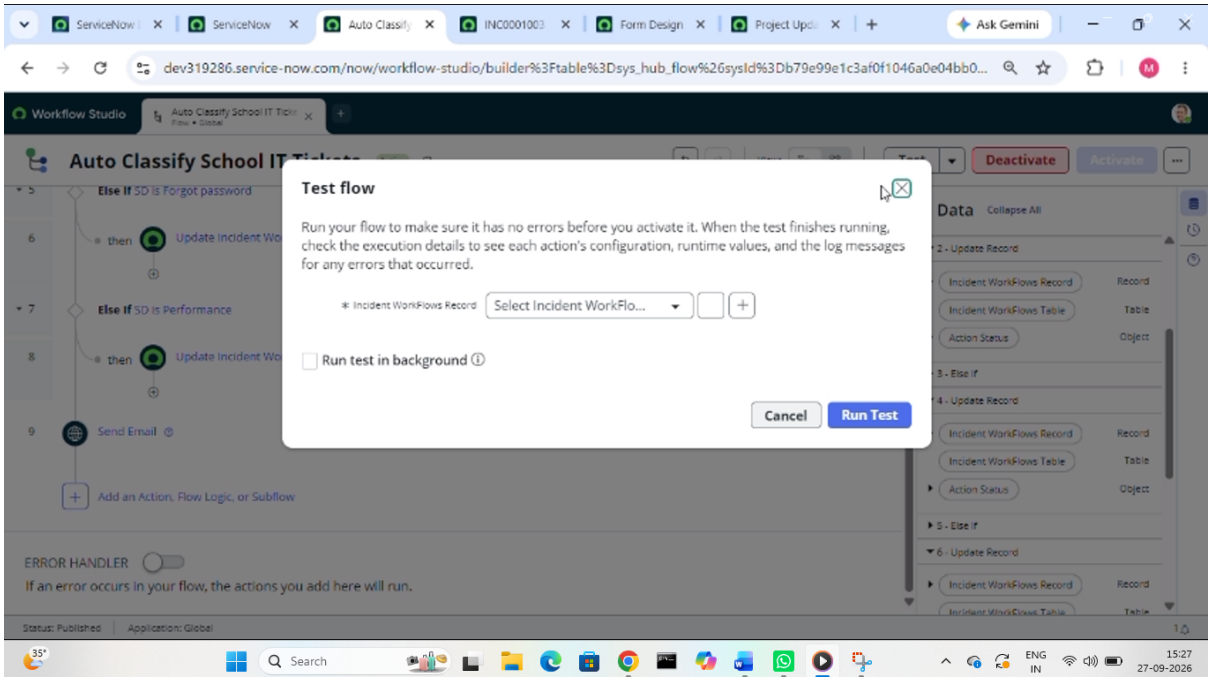
Testing summary:

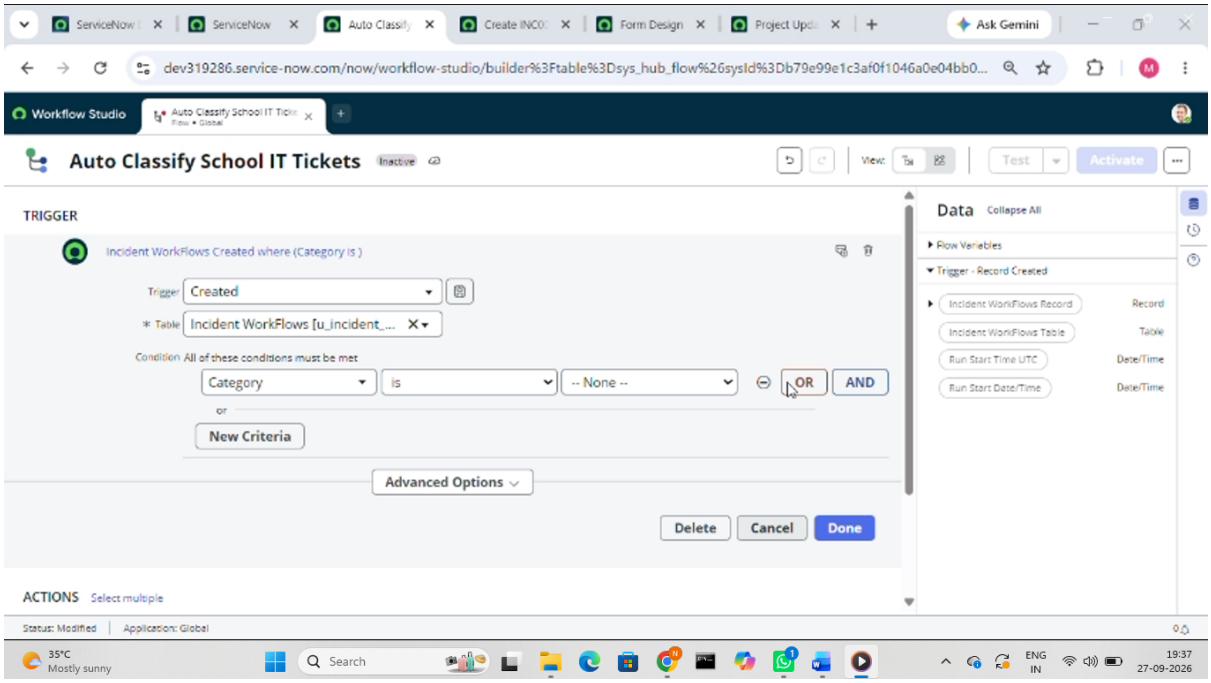
Test case	**expected result**	**status**
Wi-fi issue	network → wi-fi	pass
Projector issue	hardware → projector	pass
Password issue	access → forgot password	pass
Slow computer	performance → slow computer	pass

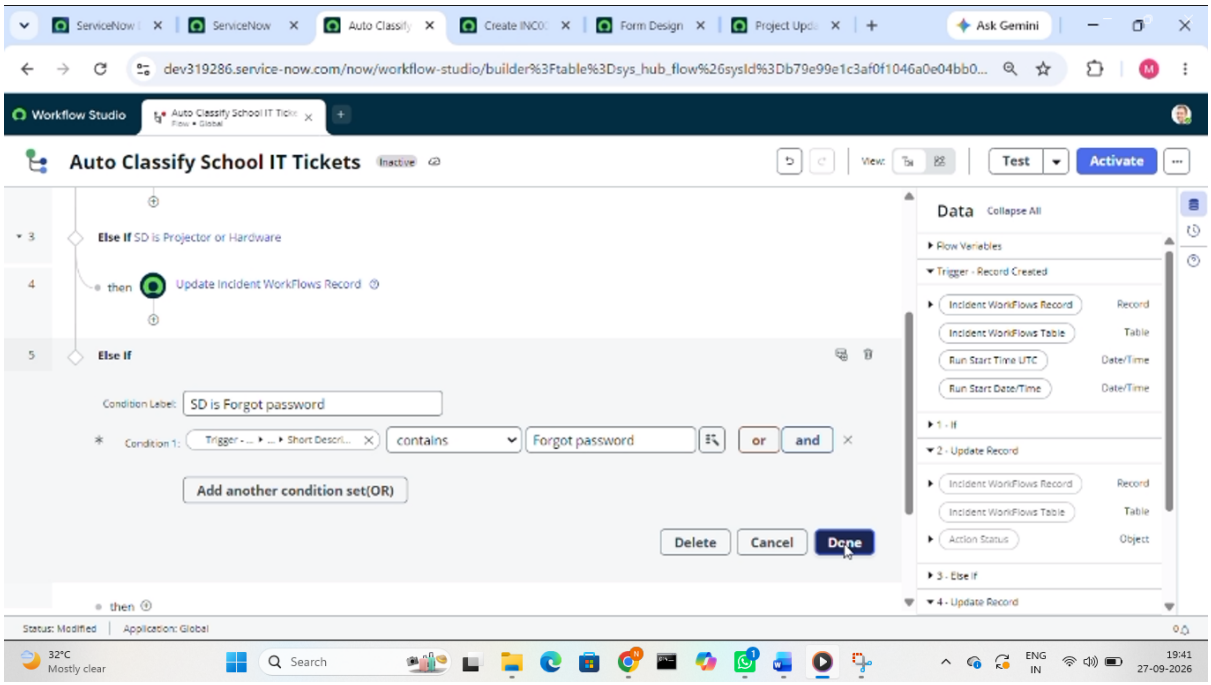
11. SCREENSHOTS

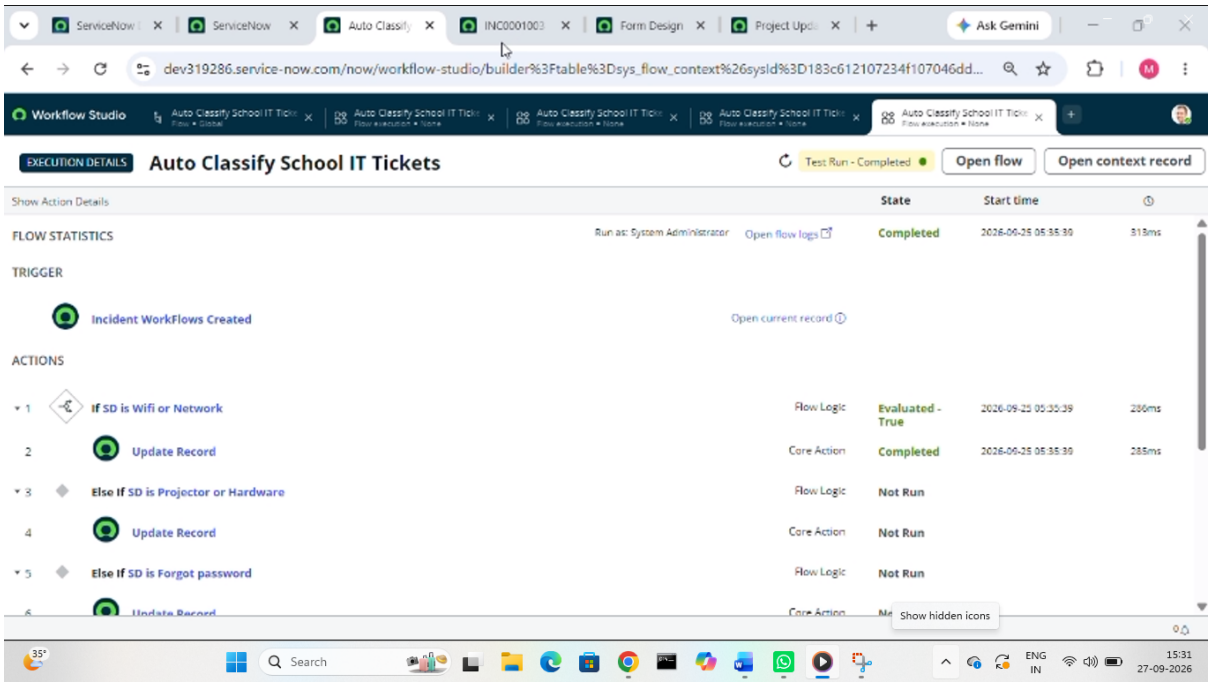





Workflow Studio
Auto Classify School IT Tickets
Deactivate
Activate
Test flow
Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.
* Incident WorkFlows Record
Select Incident WorkFlo...
Run test in background
Cancel
Run Test
Else If SD is Forgot password
Else If SD is Performance
Send Email
Add an Action, Flow Logic, or Subflow
ERROR HANDLER
If an error occurs in your flow, the actions you add here will run.
Data
Collapse All
Status: Published
Application: Global


Workflow Studio
Auto Classify School IT Tickets
Inactive
Test
Activate
TRIGGER
Incident WorkFlows Created where (Category is)
Trigger
Created
* Table
Incident WorkFlows [u_incident_...
Condition
All of these conditions must be met
Category
is
-- None --
OR
AND
or
New Criteria
Advanced Options
Delete
Cancel
Done
ACTIONS
Select multiple
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident WorkFlows Record
Incident WorkFlows Table
Run Start Time UTC
Run Start Date/Time
Status: Modified
Application: Global


Auto Classify School IT Tickets
Inactive
Test
Activate
Else If SD is Projector or Hardware
then
Update Incident WorkFlows Record
Else If
Condition Label: SD is Forgot password
Condition 1:
contains
Forgot password
or
and
Add another condition set(OR)
Delete
Cancel
Done
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident WorkFlows Record
Incident WorkFlows Table
Run Start Time UTC
Run Start Date/Time
1 - If
2 - Update Record
Action Status
3 - Else If
4 - Update Record
Status: Modified
Application: Global


EXECUTION DETAILS
Auto Classify School IT Tickets
Test Run - Completed
Open flow
Open context record
Show Action Details
State
Start time
FLOW STATISTICS
Completed
TRIGGER
Incident WorkFlows Created
Open current record
ACTIONS
1 If SD is Wifi or Network
Flow Logic
Evaluated - True
2 Update Record
Core Action
Completed
3 Else If SD is Projector or Hardware
Not Run
4 Update Record
5 Else If SD is Forgot password
Show hidden icons

12. KNOWN ISSUES

The current system has some limitations:

- Classification depends on predefined keywords.
- Unknown issue descriptions may not be classified automatically.
- New issue types require additional flow designer conditions.
- Incorrect or unclear descriptions may not match the predefined conditions.
- The system currently supports only the configured categories and subcategories.

13. FUTURE ENHANCEMENTS

The project can be extended with:

Automatic assignment of tickets to it support teams.

Sla tracking.

Advanced reporting dashboards.

Additional ticket categories.

More keyword conditions.

Predictive intelligence.

Advanced automated classification.

Final project flow:

Student / teacher

↓

Create it ticket

↓

Incident workflow table

↓

Flow designer trigger

↓

Keyword analysis

↓

Category + subcategory

↓

Update ticket

↓

Email confirmation

↓

It helpdesk